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530 PRINT : INPUT = THE DAYS ";DD$;D = VAL (DD$)
540 IF D = 0 OR D > 31 THEN PRINT "INVALID RESPONSE": PRINT : GOTO 530
550 IF M < > 2 THEN GOTO 580
560 IF (Y = 0 AND D > 28 THEN PRINT "INVALID RESPONSE": PRINT : GOTO 530
570 IF LY = 1 AND D > 29 THEN PRINT "INVALID RESPONSE": PRINT : GOTO 530
580 DX = D:MX = M:YX = Y
590 D = D + (LD / 15) / 24
600 IF (LY = 1 AND M = 2 AND INT (D) > 29) THEN M = 3:D = D - 29: GOTO 660
610 IF (LY = 0 AND M = 2 AND INT (D) > 28) THEN M = 3:D = D - 28: GOTO 660
620 IF D < 31 GOTO 660
630 IF (M = 4 OR M = 6 OR M = 9 OR M = 11) THEN M = M + 1:D = D - 30:
    GOTO 660
640 IF INT (D) > 31 THEN M = M + 1:D = D - 31
650 IF M = 13 THEN M = 1:Y = Y + 1
660 REM CALC DAYS TO DATE REQUESTED
670 REM FROM 1960,1,1 EPOCH
680 DG = 365 * Y + D + (M - 1) * 31
690 IF M > 3 THEN GOTO 730
700 REM CALC FOR JAN OR FEB
710 DG = DG + INT ((Y - 1) / 4) - INT ((Y - 1) / 100 + 1)
720 GOTO 750
730 REM CALC FOR MAR THRU DEC
740 DG = DG - INT (M * .4 + 2.3) + INT (Y / 4)
    = INT ((Y - 1) * .4 + 2.3) + INT (Y / 100 + 1)
750 NI = DG - 715875
760 GOSUB 1620: REM GET SIDEREAL TIME
770 REM INPUT OF ORBIT DATA FOR PLANETS
780 IF I = 1 THEN GOTO 950
790 RESTORE
800 DIM PD(3,9)
810 FOR YY = 0 TO 7: FOR XX = 0 TO 8
820 READ PD(YY,XX)
830 NEXT XX,YY
840 REM MERCURY
850 DATA .071477,3.8484,.388301,1.34041,.3871,.07974,2.73514
860 DATA .12223,.836013
870 REM VENUS
880 DATA .027962,3.02812,.013195,2.28638,.7233,.00506,3.85017
890 DATA .059341,1.53168
900 REM EARTH
910 DATA .017202,1.74022,.032044,1.78547,1,.017,3.33926
920 DATA 0,0
930 FOR I9 = 1 TO 3: READ PD(I9): NEXT I9
940 DATA MERCURY,VENUS,SUN
950 F = 1
960 REM CALC DATA FOR PLANETS
970 HOME
980 PRINT : PRINT : PRINT
990 PRINT "DATA REQUESTED FOR ";YX;" ";MX;" ";DX
1000 PRINT : PRINT : PRINT
1010 PRINT : J = 1
1020 FOR J = 0 TO 7: GOSUB 1230
1030 A(J) = A:D(I) = D$;L(I) = L
1040 I = I + 1: NEXT
1050 FOR I = 1 TO 2

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